Retail Revenue Optimization & Demand Forecasting

Data Science Project using Alteryx

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Project Overview
This project forecasts retail demand and evaluates pricing strategies to improve revenue performance.

Methods Used
• Time Series Forecasting
• Dynamic Pricing
• Price Elasticity Modelling
• Revenue Optimization

Retail Revenue Optimization Summary

Average Forecasted Demand (Next 6 Months): 576,190 Units

Static Pricing Revenue:$ 345,714,124.47
Dynamic Pricing Revenue:$408,908,512.00

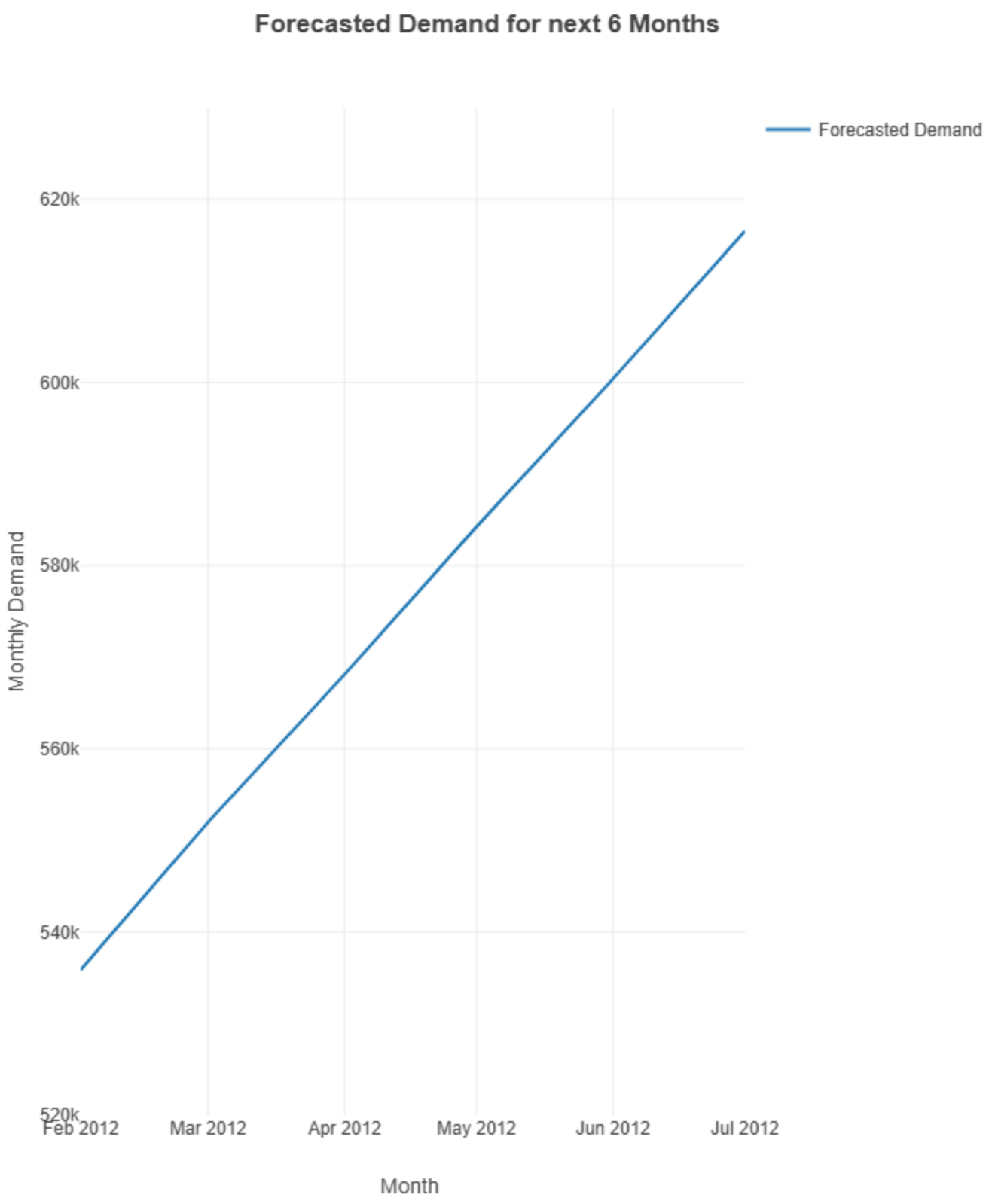
Revenue Increase from Dynamic Pricing: 18.00%

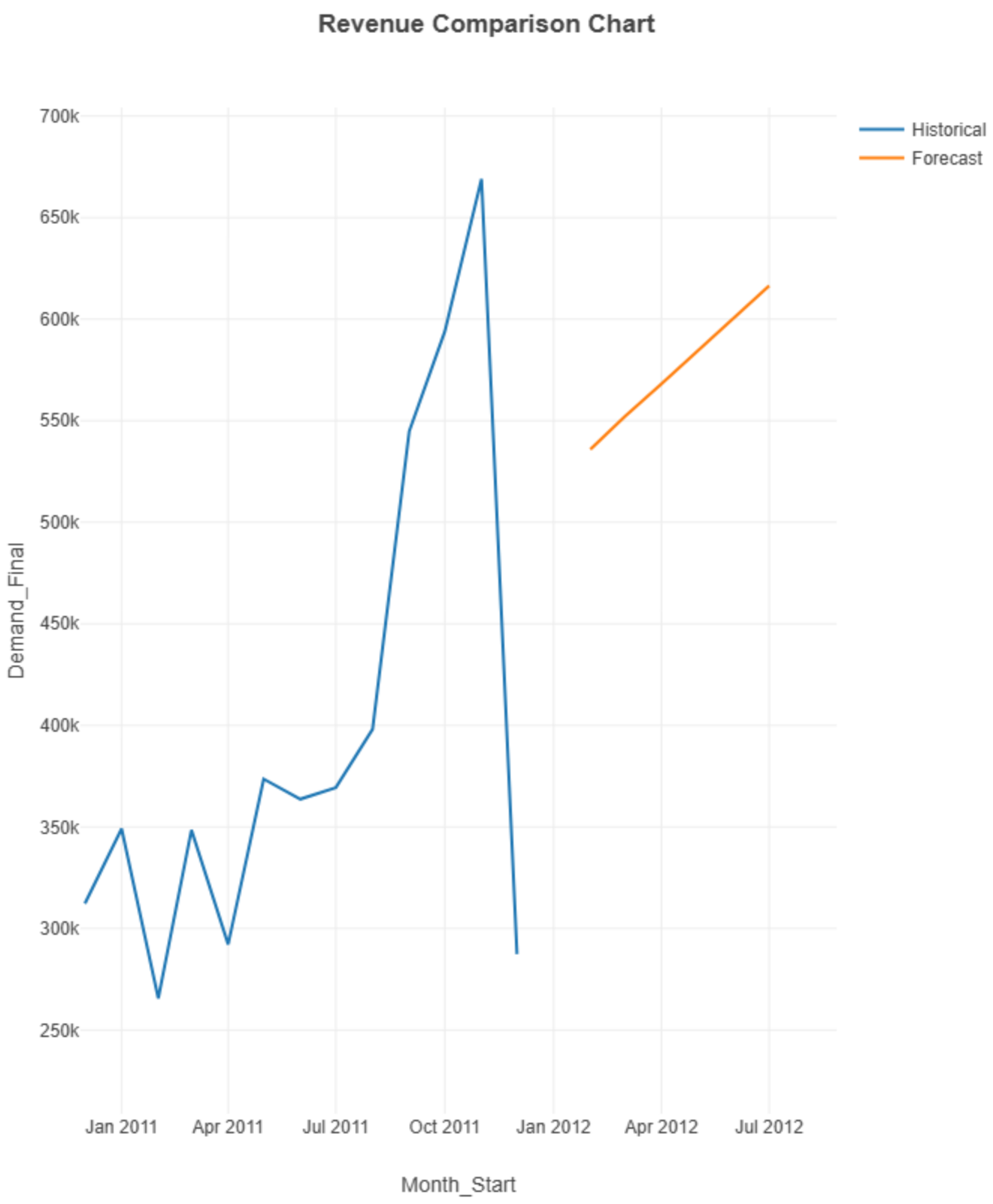
Dynamic Pricing Revenue Increase: $63,194,388

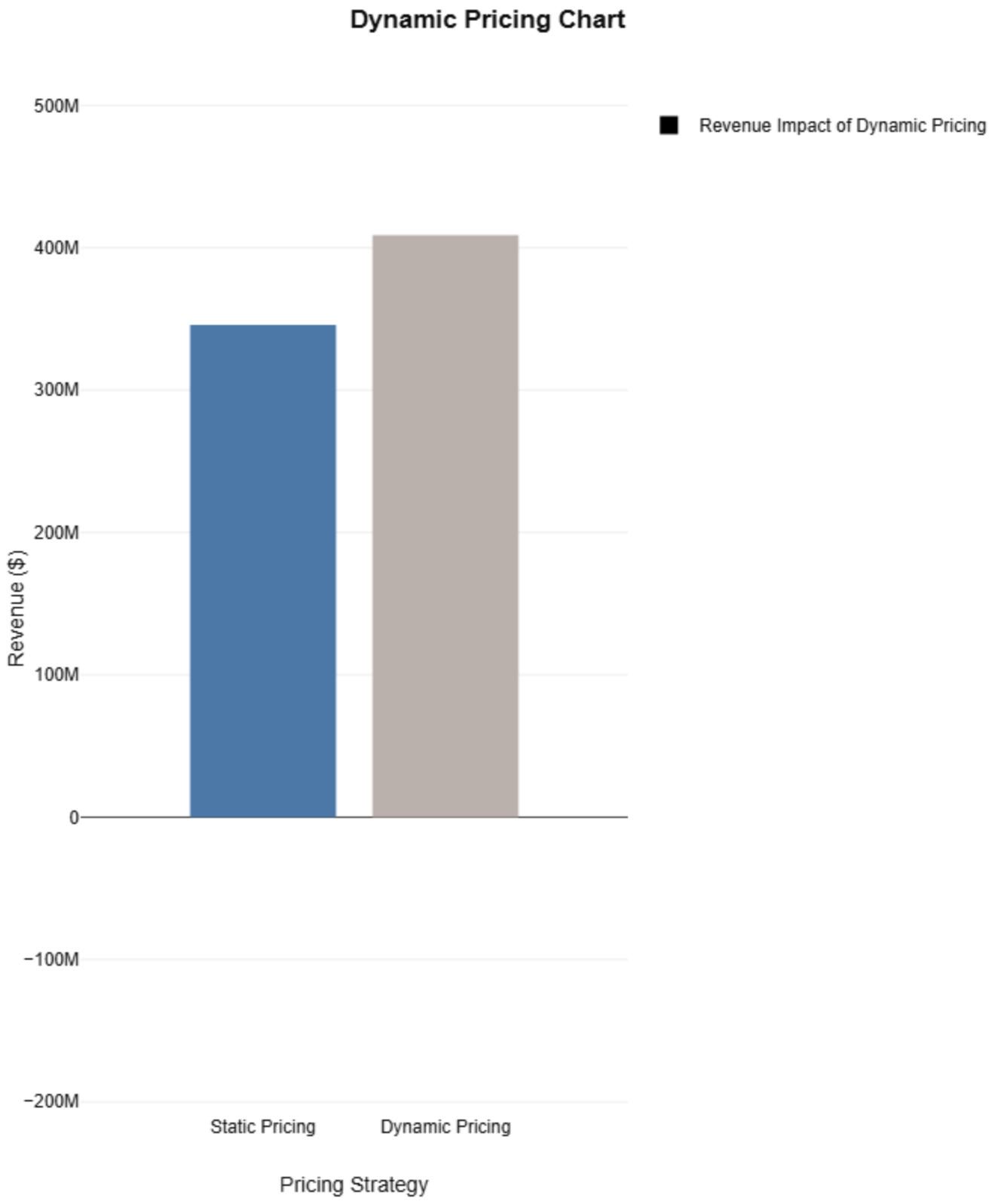
Revenue increases with higher elasticity, showing strong benefit from dynamic pricing.

Pricing Recommendation

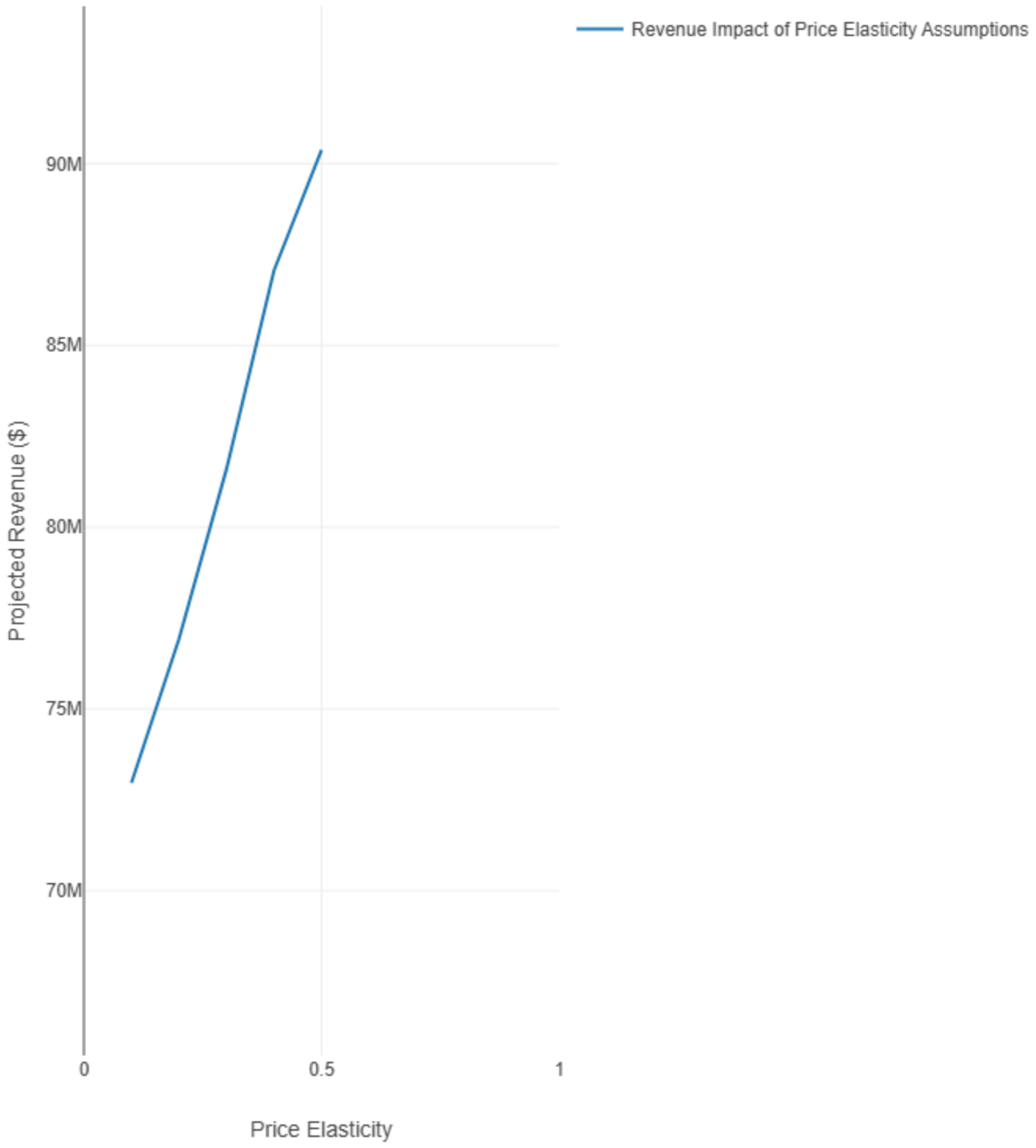
Recommended elasticity scenario: 0.5. This scenario produces projected revenue of $90M and represents the highest revenue outcome in the sensitivity analysis.


Forecasted Demand for next 6 Months
Forecasted Demand
620k
600k
580k
560k
540k
520k
Monthly Demand
Feb 2012
Mar 2012
Apr 2012
May 2012
Jun 2012
Jul 2012
Month


Revenue Comparison Chart
700k
650k
600k
550k
500k
450k
400k
350k
300k
250k
Demand_Final
Jan 2011
Apr 2011
Jul 2011
Oct 2011
Jan 2012
Apr 2012
Jul 2012
Month_Start
Historical
Forecast


Dynamic Pricing Chart
Revenue Impact of Dynamic Pricing
500M
400M
300M
200M
100M
0
−100M
−200M
Revenue ($)
Static Pricing
Dynamic Pricing
Pricing Strategy

Elasticity Sensitivity Chart



Key Insights

- Demand is expected to grow steadily based on the forecast model.
- Dynamic pricing increases total revenue by 18% compared to static pricing.
- Revenue is sensitive to price elasticity assumptions.
- Optimal pricing appears within elasticity range 0.2–0.4.